

PROBLEM STATEMENT 3 · COMPUTATIONAL PRE-CLINICAL DRUG DEVELOPMENT

ARBITER

Agentic Reasoning for Bridging Inconsistent Toxicology Evidence and Recommendation

An AI that does what a preclinical safety committee does, reproducibly and transparently: it gathers conflicting safety evidence, reasons out which evidence wins and explains why, quantifies what it cannot know, recommends the single most useful next experiment, and drafts the regulator-ready justification.

Deliverable	Solution class	Status	Date
Pitch bible and objection playbook	Neuro-symbolic, multi-agent AI	Concept + MVP roadmap	June 2026

01 Executive Summary

Ninety percent of drugs that clear preclinical testing still fail, and roughly a third of that failure is toxicity nobody reconciled in time. The industry has poured effort into *predicting* individual safety endpoints with machine learning. The unsolved bottleneck is not prediction. It is **integration**: turning a pile of conflicting signals from computational models, in-vitro assays, toxicogenomics, and organ-chips into one defensible go or no-go decision, with a justification a regulator will accept.

ARBITER automates that integration step. Its language-model agents gather the evidence; a **symbolic reasoning engine** reconciles the contradictions and shows its work; the verdict is grounded in the known biological mechanism; a planner names the one experiment that would resolve any remaining doubt; and the system auto-drafts the weight-of-evidence credibility dossier. It keeps a human in the loop at every step and never makes the final call on its own.

This sits on an unusually strong, recent regulatory tailwind. The FDA's April 2025 roadmap to reduce animal testing and its 2026 expansion of weight-of-evidence guidance are pushing the entire industry toward exactly the structured, auditable safety reasoning ARBITER produces.

WHAT IT IS

A neuro-symbolic, multi-agent system that reconciles conflicting preclinical safety evidence into a transparent, defensible decision.

WHY IT IS NEW

The pieces exist separately. No tool combines symbolic conflict-reconciliation, value-of-information experiment planning, and a regulator-facing dossier inside agent orchestration.

DATA TO BUILD IT

Entirely public for the MVP: DILIrank, ToxCast/Tox21, Open TG-GATES, AOP-Wiki, Therapeutics Data Commons, ChEMBL, CTD.

PFIZER FIT

Safety weight-of-evidence is cross-cutting: Internal Medicine, Inflammation & Immunology, Oncology, and Vaccines all need it.

THE 30-SECOND VERSION

Drug development costs one to two billion dollars per drug and fails ninety percent of the time, and a third of that is toxicity. ARBITER is an AI that reconciles the conflicting safety evidence, decides which evidence wins and explains why, and writes the verdict regulators want. It catches doomed candidates earlier and rescues borderline winners that caution would have killed, which is the most expensive decision in pharma, made faster and documented.

02 The Problem, and Why Now

The preclinical safety *decision*, not the prediction, is the expensive, unsolved part of the pipeline.

Toxicity is a primary driver of attrition

Across the industry, toxicity is estimated to account for the attrition of roughly a third of drug candidates, with each new approved drug costing on the order of one to two billion dollars by the time the failures are

paid for. The catastrophically expensive outcome is not killing a bad drug early. It is advancing a bad drug late, where a failure in Phase 2 or 3, or a post-market safety withdrawal, destroys hundreds of millions of dollars plus reputation and trust.

The bottleneck is reconciliation, not prediction

A program team routinely faces a structural-alert model saying "hepatotoxic," a 3D hepatic spheroid assay saying "clean," a toxicogenomic signature flagging a pathway, and an organ-chip readout that is ambiguous. Someone has to weigh those against each other and produce a decision plus a defensible justification. Today that synthesis is manual, slow, inconsistent between reviewers, and barely documented. It lives in meetings and in people's heads. This is precisely the weight-of-evidence task that regulators now explicitly want, and it is where ARBITER works.

The regulatory tailwind is recent and specific

The FDA released its roadmap to reduce animal testing in preclinical safety studies in April 2025, with a stated long-term aim of making animal studies the exception rather than the norm, building on the FDA Modernization Act 2.0. In April 2026 the agency reported releasing draft guidance expanding the use of weight-of-evidence approaches to support new approach methodologies, and qualifying its first AI-based drug development tool. The EMA runs a parallel qualification pathway centered on weight of evidence. A tool that *produces* structured, auditable weight-of-evidence is aligned with where regulators are actively heading.

03 The Idea: ARBITER

In plain terms: an AI judge that listens to all the conflicting drug-safety evidence, decides who is right and explains why, and writes up the verdict.

When a candidate is tested, it accumulates heterogeneous, frequently contradictory evidence of differing reliability. ARBITER takes that mess and runs it through five moves.

- 1 Gather.** Agents pull every relevant signal for the compound: computational predictions, in-vitro assay results, toxicogenomic signatures, organ-chip readouts, and the literature.
- 2 Reconcile.** A symbolic engine weighs the conflicting evidence using explicit logic rather than averaging, and explains *why* one piece outranks another.
- 3 Quantify.** It reports its confidence as an honest range, not a single false-precision number, widening that range exactly when evidence is thin or conflicting.
- 4 Ground and plan.** It maps the verdict onto the known biological mechanism, and when doubt remains it names the single most informative next experiment.
- 5 Document.** It auto-drafts the weight-of-evidence credibility dossier that internal governance and regulators expect.

What makes it different

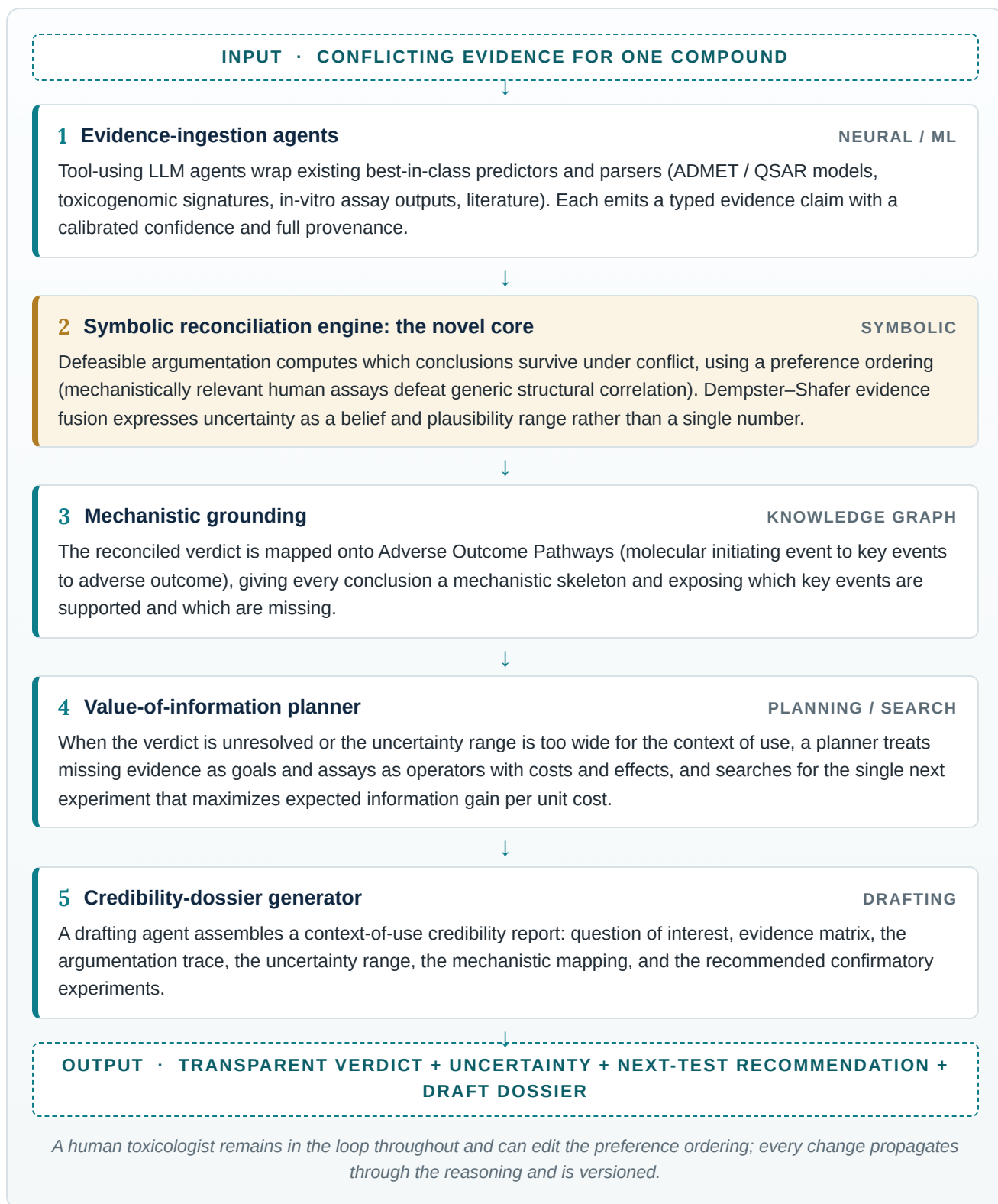
Everyone in the AI-for-drug-discovery space is building tools to *predict* toxicity or to *generate* new molecules. Nobody has built an AI to settle the conflicts between all those predictions and turn the result

into a decision you can defend. That is the whitespace. ARBITER is an integration and reasoning layer, not another predictor, which is also why it is far less data-hungry than a deep-learning model and far easier to trust.

04 How It Works: Architecture

ARBITER is a hybrid of neural perception and symbolic reasoning: machine learning reads and

predicts, and explicit logic decides and explains.



Plain-language box: what "symbolic" and "neuro-symbolic" mean

A neural network learns patterns from data into millions of numbers nobody can read. It is like someone who has seen ten thousand cases and has a gut feeling: often right, unable to explain why, and confidently wrong when it meets something new. A **symbolic** system reasons with explicit rules you can read: "this assay outranks that prediction because it measures the human mechanism directly, and the rule says mechanistic evidence beats structural correlation." You can inspect every step and challenge any rule.

Neuro-symbolic means using machine learning for the pattern-heavy parts (reading data, predicting properties) and symbolic logic for the reasoning and the decision. You get the power of ML and the transparency of logic. That transparency is the entire reason ARBITER can show its work instead of handing you a number.

05 Where the Logic Comes From

The first question a scientist asks about a rule that "one piece of evidence outweighs another" is: says who? ARBITER does not invent the weightings, and neither does the developer.

The weighting that governs which evidence wins is grounded in three sources, none of which is opinion.

1. Established evidence-hierarchy frameworks

Toxicology already has formal systems for how much to trust a study. Klimisch scores rate the reliability of a toxicology study on a standard scale used in regulatory submissions. OECD weight-of-evidence and Integrated Approaches to Testing and Assessment guidance lay out how to combine evidence types. The Adverse Outcome Pathway framework itself assigns a confidence level to each mechanistic link, based on biological plausibility and empirical support using Bradford Hill style criteria, and that confidence is curated in AOP-Wiki. ARBITER reads those existing scores. It encodes doctrine, not preference.

2. Mechanistic relevance to the human outcome

The defensible principle behind "this beats that" is closeness to the real biology. A human-cell assay measuring the specific molecular event in an adverse outcome pathway is more probative than a generic structural correlation. That ordering falls straight out of the mechanism: evidence that directly probes a key event carries more weight than evidence that only sits near it.

3. Empirical reliability calibrated on ground truth

Where ground truth exists, such as the DILIrank set of liver-injury classifications, you can measure how often each source has historically been right on a given endpoint and let those measured reliabilities tune the weighting. That is calibration from evidence, not a number anyone asserted.

What makes it defensible rather than arbitrary

The ordering is explicit, inspectable, and owned by a toxicologist, not buried in a model's weights. A scientist can open ARBITER, read "for hepatotoxicity, human hepatocyte assay outranks rat in-vivo outranks generic QSAR," and either agree or change it for their context. It is versioned, so every change is tracked. The rules are something a domain expert signs off on and can contest.

SAY THIS

ARBITER does not decide on its own which evidence wins. The weighting comes from established toxicology frameworks, from how directly each piece of evidence probes the human mechanism, and from how reliable each source has proven against known outcomes. It is fully explicit: a toxicologist can read the ordering, contest it, and edit it. The rules are governed by experts, not invented by the algorithm.

06 Uncertainty and Trust

A system that emits "95% certain" gets over-trusted and eventually burned, because biology is unpredictable. ARBITER is built to be honest about what it does not know, which is what kills the over-reliance problem.

It never reports a single false-precision number

This is the reason it uses Dempster–Shafer fusion instead of plain probability. The method gives a belief and a plausibility, and the gap between them *is* the system's ignorance made visible. Scarce or conflicting evidence produces a wide range, and a wide range is ARBITER saying, in effect, that it does not know enough to be sure. A naive model collapses that into one confident number. ARBITER refuses to.

It knows when it is off the map

Conformal prediction and applicability-domain checks detect when a compound is too different from anything the evidence covers, the exact case where biology surprises you. When that happens, ARBITER abstains and recommends a test instead of issuing a verdict. That is the correct behavior for unpredictable biology: not a confident guess, but a clear statement that it cannot vouch for this one yet.

People read an argument, not a number

A "95%" cannot be interrogated. An argument graph can. A scientist can see which evidence the system trusted and why, and push back. That transparency is the real antidote to over-reliance: the tool shows its work so a human can catch where it is wrong, which is impossible with a bare score.

The honest limit

ARBITER cannot invent signal that is not there. If a toxicity has zero precedent in any source, no system predicts it, and ARBITER will not either. What it does is report high uncertainty and route that compound to a human and a lab bench. That is the safe failure mode. The nightmare is a confident-but-wrong AI; ARBITER's design is uncertain-and-honest, which sends biology's surprises toward human scrutiny instead of past it.

07 Handling Novel Drugs

A brand-new molecule has no safety record, and no tool can look up what happens when you dose a compound that has never been dosed. The reframe is that public data is not a lookup table of old

compounds.

Public data is where the system learns how toxicity *works*, and that knowledge transfers to molecules that do not exist yet. Toxicology generalizes through mechanism and structure, not through compound identity.

Mechanism, through pathways

An Adverse Outcome Pathway is mechanism-level, not compound-level. If inhibiting a given transporter leads to cholestatic liver injury, that causal chain applies to any molecule hitting that target, including a brand-new one. For a novel compound you do not need its history; you need to know whether it triggers the known molecular initiating events, and that can be assessed through binding predictions and in-vitro assays.

Structure, through read-across and structural alerts

This is regulatory-accepted, not exotic. A novel molecule that shares a scaffold with known toxic compounds, or carries a known toxicophore, inherits those inferences. The molecule is new; the chemistry is not.

The compound generates its own evidence

A novel compound in real development is not data-less. It gets tested, producing its own in-vitro readouts, organ-chip results, and toxicogenomic signatures. That fresh, compound-specific evidence is exactly what ARBITER reconciles, read through the mechanistic and structural lens learned from public data. The public data is the lens, not the answer.

SAY THIS

Public data does not teach the system what old compounds did, it teaches it how toxicity works, mechanistically and structurally, and that transfers to new molecules. A novel drug also generates its own test data, and reconciling that against known mechanisms is exactly ARBITER's job. When the mechanism itself is new, it does not guess, it reports what it cannot know yet and names the one experiment that would answer it.

08 Data and Datasets

The entire MVP runs on free, public data. No proprietary data from any company is required to prove the concept.

Source	Role in ARBITER	Access
FDA DILrank	Ground-truth labels for drug-induced liver injury. The answer key the MVP validates against.	Free, FDA Liver Toxicity Knowledge Base
Therapeutics Data Commons	ML-ready ADMET predictions (liver tox, hERG, mutagenicity, CYP panel). The computational-prediction input.	Free, Python package

Source	Role in ARBITER	Access
EPA ToxCast / Tox21	High-throughput in-vitro bioactivity. The lab-assay evidence stream.	Free, EPA CompTox Dashboard + API
Open TG-GATES	Toxicogenomics: gene expression and pathology for a set of compounds. The omics evidence stream.	Free, public archive
AOP-Wiki	Adverse Outcome Pathways. The mechanistic knowledge graph that grounds verdicts.	Free, RDF / SPARQL
ChEMBL	Compound bioactivity and physicochemical data. Enrichment and structure linking.	Free, REST API + client
Comparative Toxicogenomics DB	Curated chemical-gene-disease interactions. Mechanistic enrichment.	Free for research

The one real engineering gotcha

Every database uses different identifiers for the same drug. The crosswalk that ties them together is chemical structure, so compounds are matched by InChIKey or SMILES (both available free via PubChem and ChEMBL). This is the thing to get working early.

Data is reference, not a giant training corpus

ARBITER is not trying to train a universal toxicity model from data that does not exist. It grounds each decision case by case: the mechanistic rules from AOP-Wiki, the assay results for the molecule in front of it, the literature pulled for that case. That is reference and lookup, which is why scarce data is far less of a constraint here than it would be for a deep-learning predictor. Where genuine conflict cases are sparse, they can be augmented by constructing adversarial evidence sets that force sources to disagree, then validating against held-out ground truth.

Hackathon scope

For the demo, three sources are enough: DILIrank for truth, Therapeutics Data Commons for predictions, ToxCast for lab signal. Pick three or four drugs where those sources genuinely disagree, and that is the whole demonstration. Everything else is for later phases.

09 Data Security and Deployment

Pharma companies will not share proprietary data, and feeding lab data into a consumer AI that stores and could leak it is a legitimate fear. Both concerns are real, and both are answered by the architecture rather than by hope.

Separate two different problems: getting data to *build* the system, and protecting data when someone *uses* it.

Building it

The MVP needs zero proprietary data; the public datasets above are sufficient. And because ARBITER's core is symbolic rather than a neural net trained on a private compound library, it does not need to ingest anyone's data to function. The system built on public data works the same inside a company on private data, without ever being trained on it.

Using it

The fear that "AI stores it and spews it out to others" is true of a free consumer chatbot. It is not how enterprises deploy AI. Self-hosted, open-weight models run entirely inside a company's own infrastructure, so data never leaves the building. Even commercial APIs offer Zero Data Retention and binding no-training contractual terms that companies at this scale already sign. And ARBITER makes this easier than most AI tools, because the sensitive numbers can flow through deterministic code and the symbolic engine, both of which run locally; the language-model agents handle orchestration and drafting, not the raw confidential data.

Precedent that the industry already solved a version of this

A consortium of ten pharmaceutical companies ran a project using federated learning in which each company contributed to a shared model without any of them seeing another's raw data: collective benefit, everything kept private. *(Verify the specifics of this program before citing it in the pitch.)*

Honest caveat

This is a deployment story, not something the hackathon MVP will prove. The demo runs on public data and will not demonstrate enterprise security. The right framing is that the architecture is built for it and the engineering pattern is known and already in use. Name it as a roadblock and show the answer is ready, rather than pretending the issue is not there.

10 Adoption and Implementation

The goal is not to convince people to trust an algorithm over what they have now. It is to show ARBITER is not an algorithm you have to trust at all: it is a transparent layer that makes the process they already trust faster and clearer.

How it earns trust

It shows its reasoning instead of asking for faith, so a scientist reads the argument and decides, the way they already evaluate a colleague. It encodes their own field's evidence standards rather than foreign logic. It is editable and expert-governed, so they own it. It admits when it does not know. And, decisively, it earns trust through **retrospective validation**: run it on historical compounds where the outcome is already known, publicly via DILrank and internally via archived programs, and show that on the conflicting cases its reasoning would have reached the right call more consistently than the ad hoc process did. That is evidence, not persuasion.

How it fits the existing workflow

It augments, it does not replace. It reads from the electronic lab notebook, the lab information system, and the assay databases already in use, and it wraps the predictors already running. Nobody changes how they capture data, and nothing is ripped out. The safety committee still makes every decision. Right now a

person hand-assembles the evidence package before a review meeting; ARBITER becomes that pre-read, structured and complete with conflicts surfaced, so the humans deliberate from a clear picture instead of a pile of disagreeing reports.

Shadow mode	Runs in parallel on real cases, output reviewed but not binding. Pure validation, zero risk, no workflow change. This is where it earns its track record.
Advisory	Once the track record holds , its dossier becomes the official pre-read that frames the safety discussion.
Integrated	It becomes part of the standard candidate-nomination package and feeds regulatory new-approach-methodology submissions. The human is the decision-maker the entire way.

Why existing teams cannot already do this

The expertise is not missing; companies have thousands of specialists and bioinformaticians. What is missing is consistency, speed, scale, and an audit trail. The reasoning today is slow (weeks of back-and-forth), inconsistent (different reviewers weigh the same evidence differently), undocumented (no structured, regulator-ready trace), and unable to scale to the exploding volume of multi-omics and high-throughput data for every compound. ARBITER encodes the experts' own standards and applies them the same way every time, with a trail regulators can read. The bioinformaticians build the predictors, which are ARBITER's inputs; ARBITER is the layer above that reconciles their conflicting outputs into a transparent decision. As for why nobody has built it yet, the enabling pieces only recently converged: mature AOP knowledge bases, agents capable enough to orchestrate, formalized argumentation, and the FDA's 2025 to 2026 push toward structured weight-of-evidence.

SAY THIS

Companies have the experts. What they do not have is a way to make that reasoning consistent, fast, scalable, and auditable. Today it lives in meetings and people's heads. ARBITER encodes the experts' own standards and applies them the same way every time, with a trail regulators can read. It is not replacing the specialists, it is the spreadsheet they never had.

11 Timeline and Cost

No false precision, which would be ironic given everything ARBITER stands for. Both are best answered as honest ranges tied to value delivered at each stage.

Timeline

Full integration into the decision process is a multi-year horizon, like any enterprise safety tool. The key point is that value does not wait for full integration: the structured evidence dossier is useful from the first month of shadow mode, so it pays off at every stage rather than all at once.

Hackathon	Working demo on public data, reconciling a handful of conflicting cases against ground truth.
3–6 months	Generalize beyond liver injury to cardiotoxicity and mutagenicity; wire in real prediction and assay pipelines; calibrate confidences.
6–12 months	Internal shadow-mode pilot on one team and one endpoint, validated against historical decisions and outcomes.
2–3 years	Advisory then integrated adoption across multiple endpoints and therapeutic areas.

Cost

ARBITER is cheap by design. It does not train an expensive foundation model from scratch; it wraps predictors that already exist. It does not need new lab infrastructure; it reads data already generated. It can run on-premise on open-weight models, so there are no per-token API bills at scale and nothing leaves the building. The real costs are engineering time and the expert time to curate and validate the rule base: operational, not capital-heavy. The MVP itself is close to free.

SAY THIS

On cost, the MVP runs on public data and open tooling for next to nothing, and because the reasoning core is symbolic it needs no expensive model training and no new lab infrastructure. Set against one late-stage toxicity failure that costs hundreds of millions, the economics are not close.

12 The Business Case

The people who fund and greenlight this do not speak the science. For this hackathon, the "investor" is Pfizer leadership deciding whether to fund building it internally. Same logic: a non-technical decision-maker asking why they should spend on this.

The discipline is translation. Investors do not buy technology, they buy returns and de-risked bets. Strip every technical term and speak in money, time, risk, and odds.

The asymmetry

It is cheap to build and set against failures that cost hundreds of millions to billions each. Move even a small fraction of kill-decisions earlier, or rescue one real winner, and the return dwarfs the cost by orders of magnitude. Tiny cost, enormous downside protection, plus upside.

It is fundamentally a risk-reduction play

It lowers the odds of the two events that tank a pharmaceutical company: the catastrophic Phase 3 failure and the post-launch safety withdrawal. It throws off an audit trail that cuts legal and regulatory liability. The regulatory tailwind tells an investor adoption is coming whether or not they bet, so they would rather bet early.

The incentive reframe: catching problems early makes money

The assumption that killing drugs loses money is backwards over any real horizon. The expensive mistake is advancing a doomed drug for years. Killing it early does not lose money; it stops money from being

burned and frees it for candidates that can win. And ARBITER is not a kill switch: it also greenlights safe candidates with confidence, and when one is borderline it names the single test that de-risks it, rescuing winners a conservative manual process would have killed out of caution. Accountability stays human, so no executive is asked to bet the company on an algorithm.

SAY THIS (THE INVESTOR VERSION)

Every drug that fails late costs this company hundreds of millions, and a third of those failures are safety. ARBITER catches the bad candidates earlier and saves the good ones that caution would have killed, the most expensive call in your pipeline, made faster and documented. It runs on data and infrastructure you already have, and it lines up with where the FDA is already pushing the whole industry. The cost is a rounding error against a single avoided failure.

13 Pfizer Fit

Safety weight-of-evidence is not a niche; it is a cross-cutting capability every therapeutic area depends on.

A reconciliation-and-decision layer for preclinical safety touches Internal Medicine, Inflammation & Immunology, Oncology, and Vaccines alike, because every program has to clear safety before it advances. It slots into Pfizer R&D's existing candidate-nomination governance as the structured pre-read for the safety review, and it produces exactly the auditable weight-of-evidence artifacts the FDA's 2025 to 2026 direction is pushing the industry toward. That gives Pfizer both an internal efficiency gain and a regulatory and speed advantage as new-approach-methodology submissions become more common. The fact that the preclinical stage carries a lighter regulatory burden than clinical work, and that ARBITER produces regulatory artifacts rather than making autonomous medical decisions, makes it a low-friction place to deploy advanced AI.

14 The 10/10 Scorecard

Benchmarked against Pfizer's five judging criteria, with the weakness and mitigation named for each.

Criterion	Score	Why it hits, and how to defend it
Innovation	10	First system to put symbolic defeasible-argumentation and Dempster–Shafer reconciliation and a value-of-information planner inside multi-agent orchestration for preclinical safety, with an auto-drafted dossier. <i>Mitigation:</i> differentiate explicitly from adjacent agentic-toxicology work, which orchestrates tools but does not do formal reasoning, experiment planning, or a credibility dossier.
Pfizer Fit	10	Cross-cutting across all four portfolio areas; slots into candidate-nomination governance; aligned with the FDA roadmap. <i>Mitigation:</i> name a concrete first endpoint and therapeutic area in the pitch.
Technical Feasibility	10	MVP buildable on free public data; the neural layer reuses off-the-shelf predictors; the symbolic layer uses established formalisms. DILrank gives

Criterion	Score	Why it hits, and how to defend it
		clean ground truth. <i>Mitigation:</i> if predictor calibration is weak, use a curated hand-built conflict set for the demo.
Non-Technical Feasibility	10	Lighter preclinical regulatory burden; produces regulatory artifacts rather than autonomous decisions; aligned with FDA and EMA direction; strong ROI. <i>Mitigation:</i> frame the credibility-dossier standard as aligned with direction, not as already accepted.
Presentation	10	A single dramatic demo: a conflict where averaging is wrong and ARBITER's readable argument is right. Citable primary sources. Symbolic transparency makes it explainable to a non-technical panel. <i>Mitigation:</i> rehearse the plain-language version alongside the technical one.

15 What the Video Must Cover

A crosswalk so the five-minute pitch hits every element Pfizer's brief requires.

Required element	Where it lives in this document
Your proposed solution	Sections 3 and 4: the idea and the architecture, plus the 30-second version in Section 1.
Your roadmap to an MVP	Sections 8 and 11: hackathon scope and the staged timeline.
Alignment with Pfizer	Sections 2 and 13: the regulatory tailwind and cross-portfolio fit.
Key challenges and roadblocks	Sections 9, 10, and 17: data, adoption, and consolidated challenges.
Expected outcome	Section 18: outcomes and impact metrics.

16 Objection-Handling Playbook

Every hard question raised in the stress-test, with a tight answer and a ready-to-say line. This is the section to study before the Q&A.

Would you be using this to replace human revision?

No. It is human-in-the-loop by design. The toxicologist edits the weighting, reviews the verdict, and overrides anything. The whole reason the reasoning is symbolic and traceable is so a human can scrutinize every step. It solidifies the arguments researchers make; it does not replace their judgment.

SAY THIS

It is a copilot that organizes the evidence and surfaces the reasoning so human and regulatory review is faster and better grounded. The final call always stays with people.

Isn't slow consensus there for good reason? Years of lab work and failures?

Yes, and ARBITER does not make researchers agree faster. It makes their disagreement legible. It surfaces where and why the evidence conflicts, puts every assumption on the table, and handles the tedious bookkeeping. The human debate gets sharper, not removed. You are clearing the wasted motion around the rigor, not the rigor itself.

Pharma won't share data, and AI stores it and could leak it.

Both true and both answered by architecture, not hope. The MVP needs no proprietary data. The symbolic core is not trained on a private library, so it does not need anyone's data to work. In deployment, self-hosted open-weight models keep data inside the building, and the sensitive numbers never need to touch the language model. See Section 9.

SAY THIS

ARBITER runs inside a company's own secure environment. Proprietary data never leaves it and is never used to train any external model.

Won't a "95% certain" output get over-trusted and burned by unpredictable biology?

It never emits a single confident number. Dempster–Shafer gives a belief-to-plausibility range whose width is the system's ignorance made visible. It abstains when a compound is off-distribution and recommends a test instead. And people read its argument, not a score, so they can catch where it is wrong. See Section 6.

There's not much data to train on.

Correct, and that is the argument for this approach, not against it. ARBITER's reasoning core is not trained on scarce data; it reasons over whatever evidence exists case by case. With mountains of clean data you would just train a model. It is precisely the sparse, conflicting, low-signal world of real toxicology where careful reasoning plus honest uncertainty beats one overconfident net.

Where does the logic that one piece outweighs another come from?

From established toxicology frameworks (Klimisch reliability scores, OECD weight-of-evidence and IATA, AOP key-event confidence), from how directly evidence probes the human mechanism, and from reliability calibrated against known outcomes. It is explicit and expert-editable. See Section 5.

SAY THIS

The rules are governed by experts, not invented by the algorithm.

How do you convince people to trust an algorithm, and realistically implement it?

You do not ask for trust on faith; you prove it retrospectively against known outcomes, and you deploy in shadow mode first so the team watches its track record before it is ever binding. It augments the existing committee and plugs into existing systems. See Section 10.

SAY THIS

It deploys in shadow mode first, alongside the current safety review without changing it. The committee still makes every call.

How long, and how much?

Full integration is a multi-year, staged rollout, but value lands from the first month of shadow mode. Cost is low by design: no foundation-model training, no new lab infrastructure, runs on-premise on open models. Set against a single late-stage failure costing hundreds of millions, the economics are not close. See Sections 11 and 12.

What if you kill a future billion-dollar drug? What will CEOs think?

ARBITER does not kill anything; people do, and its transparency makes a wrong call catchable rather than hidden. There is no universal "unsafe"; the risk threshold is the program's and the regulator's, and ARBITER takes it as input. It actively rescues borderline candidates by naming the de-risking test. What scares a CEO is a Phase 3 blowup or a post-launch withdrawal, and this reduces exactly those, with a documented trail for every decision.

How is public data useful for novel drugs with no data?

Public data teaches how toxicity works, mechanistically and structurally, and that transfers to new molecules. The novel compound also generates its own test data, which is what ARBITER reconciles. When the mechanism itself is new, it reports what it cannot know and names the experiment that would answer it. See Section 7.

Why can't existing teams with thousands of specialists already do this?

They do it, by hand, and that is the problem. What is missing is consistency, speed, scale, and an audit trail. ARBITER encodes the experts' own standards and applies them the same way every time. The specialists build the predictors; ARBITER reconciles their conflicting outputs. See Section 10.

How do you convince a non-technical investor who knows nothing about medicine?

Strip the science and speak in money, time, risk, and odds. Tiny cost, catastrophic-failure prevention, regulatory tailwind. Lead with the business case and keep the science as the credibility that survives due diligence. See Section 12.

17 Key Challenges and Mitigations

Challenge	Mitigation
Confidence calibration of upstream predictors	Use conformal prediction and per-source reliability priors learned from ground truth; the fusion layer degrades gracefully to a wide range rather than failing silently.
Legitimacy of the preference ordering	Elicit it from toxicologists, ground it in published frameworks, and version every change. It is inspectable and contestable, not hard-coded opinion.
Scientist trust in automation	Full symbolic traceability, human-in-the-loop editing, and retrospective validation against known outcomes before anything is binding.
Regulatory novelty of the dossier standard	Frame the credibility dossier as aligned with the FDA's 2026 weight-of-evidence direction, not as an already-standardized pathway.
Sparsity of genuine conflict cases	Augment with constructed adversarial evidence sets and validate against held-out ground-truth labels.
Truly first-in-class mechanisms with no precedent	The hardest case for any method. ARBITER leans on the compound's own generated data and reports wide uncertainty rather than a false verdict.

18 Expected Outcomes and Impact

What success looks like, and how to measure it.

ARBITER delivers faster, reproducible, auditable safety go and no-go decisions; fewer late toxicity failures, which are the costliest kind; and direct support for animal-testing reduction and new-approach-methodology submissions, aligned with FDA and EMA direction. It is measurable against concrete metrics rather than vague promises.

- **Accuracy on the conflicting-evidence subset** versus a naive majority-vote ensemble baseline. This is the headline demo metric.
- **Reduction in confirmatory assays** required, driven by the value-of-information planner recommending only the experiment that matters.
- **Time to a complete weight-of-evidence dossier**, versus manual assembly.
- **Inter-reviewer consistency**: how much variation between reviewers the structured process removes.

19 Sources and Further Reading

Citing sources adds credibility, which Pfizer explicitly encouraged. Verify each before quoting it in the pitch.

1. FDA, "Roadmap to Reducing Animal Testing in Preclinical Safety Studies" (April 2025), and the FDA news release on year-one progress, including expanded weight-of-evidence draft guidance (April 2026).
2. Sun et al., "Why 90% of clinical drug development fails and how to improve it?", *Acta Pharmaceutica Sinica B* (2022): attrition and failure-cause breakdown.
3. Park, Ogunseitan & Lejano (2014): Dempster–Shafer theory applied to a regulatory decision process for selecting safer alternatives to toxic chemicals, the precedent for evidence fusion in regulatory toxicology.
4. OECD, Adverse Outcome Pathway development handbook and the AOP Knowledge Base (AOP-Wiki): key-event-relationship confidence and weight-of-evidence assessment.
5. Klimisch et al. (1997): the standard system for evaluating the reliability of toxicological data.
6. FDA DILIrank (Liver Toxicity Knowledge Base) and the Therapeutics Data Commons ADMET benchmark: the public data backbone.
7. *npj Digital Medicine* (2025), "AI-driven virtual cell models in preclinical research": context on AI, new-approach-methodologies, and the FDA Modernization Act 2.0.
8. EPA ToxCast and Tox21 program documentation, and Igarashi et al. (2014) for Open TG-GATES: in-vitro bioactivity and toxicogenomics data.

20 Caveats and Assumptions

Stated plainly, because intellectual honesty is part of what makes the pitch credible.

- The agentic-toxicology space is moving quickly. Claim novelty *as of mid-2026* and emphasize ARBITER's specific differentiators (symbolic reconciliation, value-of-information planning, the credibility dossier) rather than "first agent in toxicology."
- Applying a formal model-credibility standard to drug-safety weight-of-evidence dossiers is an emerging extension without a standardized regulatory precedent. Frame it as aligned with direction.
- Some referenced multi-agent and federated-learning results are early or preprint-stage. Present them as landscape, not settled fact, and verify before citing.
- Public dataset sizes cited here are point-in-time and will continue to grow.
- The biggest technical risks to a clean demo are upstream predictor calibration and the sparsity of genuine conflict cases; de-risk both early.

One closing reminder for the team. The technology is the easy part. Adoption is the real work, and it is slow in pharma for good reasons. What makes ARBITER realistic is that it never demands a leap of faith: it earns its way in through shadow-mode evidence, and it fits the existing process instead of fighting it.